VOLODYMYR POLISHCHUK

Head of Software & Autonomy | Systems Architect | GPS-Denied UAV Navigation

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SUMMARY

Head of software and autonomy and systems architect with 19 years of designing production infrastructure at scale: Wix's serverless platform for about 3,000 developers, Playtech's real-time betting engine at 1M requests per second, and UkrCard's payment network. Since 2026 I have built the software and autonomy function of Applied Technology Collective, a Ukrainian UAV developer, from zero: GPS-denied navigation now in production on a $15 companion computer, with its own flight-controller firmware and simulator, and the team behind it. I work from first principles, breaking a problem down to its physical, mathematical and computational realities instead of adopting frameworks by analogy. I build AI-native, owning the architecture, the mathematics and the acceptance criteria and directing AI coding agents as the implementation team.

EXPERIENCE

Applied Technology Collective | Head of Software & Autonomy

Kyiv, Ukraine | 2026 – Present

Ukrainian developer of strike, loitering-munition and cargo UAVs. Joined as the only software engineer, built the software and autonomy function from zero and hired its first two engineers. NADIR, SEXTANT and ORRERY are my own IP, used by the company; code walkthrough on request.

NADIR — GPS-Denied Position Hold (In Production)

- Freed the company's drones from depending on GPS to hold position with a visual-inertial navigation system: one downward camera and the autopilot's IMU give the velocity and position ArduPilot flies on in place of satellites.
- Kept the hardware to $15 per drone by running the whole system on a Raspberry Pi Zero 2 W, shipped as a sealed product that operators calibrate in the field and that refuses to arm until the setup is safe.

SEXTANT — Long-Range Terrain Navigation (Rolling Out)

- Extended GPS-denied flight from holding position to navigating over long range by fixing absolute position and heading from the terrain itself: the forward camera's skyline is matched against a terrain model, and downward motion stereo reads the relief below.
- Chose terrain geometry over image matching so that navigation survives snow, foliage and changing light, the conditions that defeat appearance-based systems.

ORRERY — Standalone Flight Simulator

- Cut the cost and risk of flight testing with the simulator in which NADIR and SEXTANT pass their acceptance tests before reaching an aircraft, coupled to ArduPilot SITL with cameras modelled as real sensors.
- Made it independent of our own stack: any ArduPilot SITL build, any camera and any vehicle plug in through configuration alone.

Wix.com | Principal Software Engineer / Architect

Kyiv, Ukraine | 2017 – 2026

Platformization, API Governance & Engineering Culture

- Laid the technical foundation for opening Wix's internal APIs to third-party developers and the Wix App Market by leading the company-wide adoption of gRPC and Protobuf as the API contract.
- Standardized API design across the engineering organization by building a Protobuf linter into the core CI/CD pipeline that every service passes, with code and documentation generated from the contract.
- Moved the backend to asynchronous code and cut service resource costs by designing and teaching the asynchronous-programming courses and workshops, mandatory for Wix's 1,000+ server engineers.
- Shaped Wix's engineering hiring through 300+ interviews across all three stages of the loop and a vote on the hiring committees, which also set each new hire's career path.

Internal Serverless Platform (App Engine)

- Took Kubernetes, service wiring and CI/CD off the hands of about 3,000 developers, mostly front-end, across 2,000+ microservices by architecting Wix's internal serverless platform and building it with a core team of five; control plane in Scala and ZIO with typed errors.
- Delivered zero-downtime scaling with a throughput-based autoscaler that goes beyond KEDA/HPA: it pre-warms capacity before datacenter switches and adjusts CPU and memory live.

AI Chat Agent, Domains & Core Infrastructure

- Brought AI assistance to the Wix App Market by architecting its chat agent: retrieval-augmented generation over Vespa, and a bidirectional streaming pipeline from WebSockets to gRPC for real-time LLM responses.
- Made Wix independent of third-party registrars by architecting its ICANN-accredited registrar platform and an in-house DNS server, took SSL for 8M+ domains out of human hands with automated issuance and renewal, and led the migration of the legacy Java domain-management monolith to microservices.

Playtech | Software Architect / Tech Lead

Kyiv, Ukraine | 2012 – 2017

- As tech lead of a team of five (a team leader handled management), set the design, technical direction and goals, and led the sports-betting platform's migration from a Java EE monolith to reactive Scala (Scalaz, algebraic domain modeling).
- Held 60M requests/min peaks during the FIFA World Cup with real-time streaming over WebSockets and raw sockets, kept within bounded memory by back-pressured Akka Streams.
- Cut bare-metal hardware costs by shrinking the deployment to a highly available 3-node minimum, observed through New Relic.
- Kept the platform within regulation that ruled out the cloud by designing a Dynamo-style, eventually consistent key-value store on LevelDB over bare-metal disks, with reads served locally and no network round-trip.

UkrCard | Systems Architect

Kyiv, Ukraine | 2007 – 2012

- Secured the payment network against interception by leading the move of every ATM, POS and self-service terminal from wired X.25 to wireless TCP/IP with end-to-end VPN and SSL encryption (2008–2012).
- Cut costly false positives in fraud detection with a neural network written from scratch in Java and a human-in-the-loop meta-model that learned from operator feedback (Master's project, 2007–2008).

SKILLS

Autonomy: visual and visual-inertial odometry, EKF sensor fusion, camera calibration and timing, terrain-referenced navigation, ArduPilot (EKF3, firmware, SITL), MAVLink.
Languages: Rust, C, Scala (ZIO, Akka, Scalaz), Java, Python, Node.js.
Systems: embedded Linux on ARM, real-time pipelines, Kubernetes, gRPC/Protobuf, distributed storage, CI/CD, Godot.
AI: RAG, LLM applications, neural networks from first principles.

EDUCATION

M.Sc. in Cybernetics | Taras Shevchenko National University of Kyiv, Faculty of Cybernetics. Thesis: Ensuring the Dependability of Transaction Classification in a Payment System.

B.Sc. in Applied Mathematics | Taras Shevchenko National University of Kyiv, Faculty of Cybernetics

MILITARY & OTHER QUALIFICATIONS

Armed Forces of Ukraine (reserve): Junior Lieutenant, military specialty 708200: operation, tactical deployment and maintenance of air-defense radar systems.
IYT Skipper (International Yacht Training): practical navigation.
Interests: world models for robot navigation (V-JEPA, Genesis); basal cognition (Michael Levin); foundations of physics.